Lab 2: Database Design and Normalization

VUA-SEN307 Database Systems — Week 4 Practical

1. Unnormalized table

The original sheet stores several course codes, titles and lecturers in single cells. This creates repeating groups and makes updates difficult.

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems; Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng.; Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems; Operating Systems	Dr A. Okafor; Mrs F. Eze	Software Eng.; Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

2. First Normal Form (1NF)

Each cell contains one value and each student-course registration has its own row. The composite key is (matric_no, course_code, session).

matric_no	student_name	student_dept	course_code	course_title	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	2026/2027

What was fixed: repeating lists were split into single values and each student-course registration was placed on its own row.

3. Second Normal Form (2NF)

Student names and departments depend on matric_no, while course titles depend on course_code. These details are separated from Enrolment to remove partial dependencies.

matric_no (PK)	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

course_code (PK)	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

matric_no (PK, FK)	course_code (PK, FK)	session (PK)
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

What was fixed: attributes that depend on only part of the composite key were moved into separate tables.

4. Third Normal Form (3NF)

Department details are stored once in Department, and Student and Lecturer link to it using dept_id. CourseLecturer records which lecturer teaches a course in a particular session.

Table	Columns and keys
Department	dept_id INT PK; dept_name VARCHAR(100) UNIQUE NOT NULL
Student	student_id INT PK; matric_no VARCHAR(20) UNIQUE NOT NULL; full_name VARCHAR(100) NOT NULL; dept_id INT FK → Departm
Lecturer	lecturer_id INT PK; lecturer_name VARCHAR(100) NOT NULL; dept_id INT FK → Department.dept_id
Course	course_id INT PK; course_code VARCHAR(20) UNIQUE NOT NULL; course_title VARCHAR(100) NOT NULL
Enrolment	student_id INT PK/FK → Student; course_id INT PK/FK → Course; session VARCHAR(20) PK
CourseLecturer	course_id INT PK/FK → Course; lecturer_id INT PK/FK → Lecturer; session VARCHAR(20) PK

What was fixed: department and lecturer details were separated so non-key attributes depend on the key, the whole key and nothing but the key.

5. Relationships and cardinalities

Relationship	Cardinality
Department → Student	One department has many students; each student belongs to one department (1:M).
Department → Lecturer	One department may have many lecturers; each lecturer belongs to one department (1:M).
Student ↔ Course	Many-to-many (M:N), resolved by Enrolment.
Course ↔ Lecturer	Many-to-many (M:N), resolved by CourseLecturer; session is recorded for each teaching assignment.